ARBITRARY POINTED MAPS BETWEEN POSTNIKOV TRIPLES, AND THE SPHERES

Abstract. In homotopy type theory, pointed simply connected 3-types are classified by triples (A, B, k) with $k : K(A, 2) \to_* K(B, 4)$. We show that *arbitrary* pointed maps between their realisations are classified as well: the canonical map from triples of a homomorphism a, a homomorphism b and a pointed homotopy $K(a, 2)\,;k' = k\,;K(b, 4)$ to pointed maps of fibres is an equivalence of types. It follows that the homotopy precategory of triples is isomorphic to that of pointed simply connected 3-types. We then compute the universal properties of S^2 and, through the looping embedding of pointed 2-connected 4-types, of S^3 on the side of triples, and recover $\pi_3(S^2) \cong \mathbb{Z}$ and $\pi_4(S^3) \cong \mathbb{Z}/2$. All results are checked formally; this version omits the correspondence with the formal declarations, which is in the companion note.

Attribution. The strategy (classify objects and arbitrary maps, identify the sphere universal objects on the side of triples, and treat $\pi_4(S^3)$ through the looping embedding) and the theory of deloopings used below are David Wärn's. The classification of objects is Hart–Milner's (Owen Milner's equivalence). We do not know Owen Milner's unpublished treatment of maps and do not claim to reproduce it.

Conventions. A type is *n-connected* or an *n-type* in the usual (homotopical) indexing. Composition is diagrammatic: $f\,;g$ is "f, then g"; for pointed maps $(f\,;g)_1 = g_1 \circ f_1$ and $(f\,;g)_0 = \mathsf{ap}_{g_1}(f_0) \cdot g_0$. Path composition $p \cdot q$ is defined by induction on q, so $p \cdot \mathsf{refl} = p$ holds by definition. $K(A, n)$ denotes the Eilenberg–MacLane space and $K(h, n)$ the map induced by a homomorphism h.

1. The statement

A *triple* is $t = (A, B, k)$ with abelian groups A, B and a pointed map $k : K(A, 2) \to_* K(B, 4)$. Its realisation is the fibre

$$\mathrm{Fib}k = \textstyle\sum_{x:K(A,2)} (k\,x = \mathrm{pt}),$$

pointed by (pt, k_0); it is a pointed simply connected 3-type, and $\mathsf{realise} : \mathsf{Triples} \to \mathsf{SC}_3$ is an equivalence of types (Hart–Milner). Fix triples $t = (A, B, k)$, $t' = (A', B', k')$ and write $V = K(A, 2)$, $Y = K(B, 4)$, $V' = K(A', 2)$, $Y' = K(B', 4)$, $P_x = (k_1 x = y_0)$, and $p : \mathrm{Fib}k \to_* V$ for the projection. We use: V is simply connected and a 2-type, Y and Y' are 3-connected 4-types, so each P_x is 2-connected and each path type of Y' is a 3-type.

Definition 1.

$$\mathsf{Mor}(t, t') = \sum_{a:\mathrm{Hom}(A,A')} \sum_{b:\mathrm{Hom}(B,B')} \big(K(a, 2)\,;k' =_{\mathrm{Map}_*(V,Y')} k\,;K(b, 4)\big).$$

The third component is a *path of pointed maps*: a homotopy together with its coherence at the base point.

Definition 2. For (a, b, H) put $u = K(a, 2)$, $v = K(b, 4)$, let $hs_x : k'_1(u_1 x) = v_1(k_1 x)$ be the underlying homotopy of H and $c : \mathsf{ap}_{k'_1}(u_0) \cdot k'_0 = hs_{v_0} \cdot (\mathsf{ap}_{v_1} k_0 \cdot v_0)$ its coherence. The *canonical map* $\rho(a, b, H) : \mathrm{Fib}k \to_* \mathrm{Fib}k'$ is

$$(x, q) \longmapsto \big(u_1 x,\ hs_x \cdot (\mathsf{ap}_{v_1} q \cdot v_0)\big),$$

Date: 28 October 2026.

pointed by the pair of u_0 and the path obtained from c.

Theorem 3. *For all triples t, t', the canonical map $\rho : \mathsf{Mor}(t,t') \to \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$ is an equivalence of types.*

No hypothesis on the groups or on k, k' is needed.

Remark 4. (1) Both sides are 1-types and, in general, not sets: $\mathrm{Map}_*(V, Y')$ is a 2-type, so its path types are 1-types; $\mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$ is a 1-type since $\mathrm{Fib}k$ is simply connected and $\mathrm{Fib}k'$ is a 3-type.

(2) The homotopy is data. For $t = (A,0,0)$, $t' = (0,C,0)$ the coefficients are trivial and both sides are $\mathrm{Map}_*(K(A,2), K(C,3))$, whose set of components is $H^3(K(A,2); C)$, non-zero in general.

(3) Classically the theorem follows from a connectivity estimate for the cofibre of $\mathrm{Fib}k \to K(A,2)$. We use no cofibres; the key lemma below is a direct fibrewise replacement of that estimate.

2. Proof of Theorem 3

For a pointed $c = (c_1, c_0) : V \to_* Y'$ put $Q^c_x = (c_1 x = y'_0)$ and

$$T(c) = \sum_{F : \prod_x P_x \to Q^c_x} \big(F_{v_0} k_0 = c_0\big), \qquad S(c) = \sum_{\beta : \mathrm{Map}_*(Y,Y')} \big(c =_{\mathrm{Map}_*(V,Y')} k \,;\, \beta\big).$$

$T(c)$ is the type of pointed null-homotopies of $p\,;c$ in fibrewise form, and $S(c)$ the type of factorisations of c through k.

2.1. Maps out of $\mathrm{Fib}k$ into $\mathrm{Fib}k'$.

Lemma 5. *The map* $\mathrm{Dec} : \sum_{\alpha : \mathrm{Map}_*(V,V')} T(\alpha\,;k') \to \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k')$,

$$(\alpha, F, e) \longmapsto \big((x,q) \mapsto (\alpha_1 x, F_x q),\ (\alpha_0, \pi(e))\big),$$

is an equivalence. Here $e : F_{v_0} k_0 = \mathsf{ap}_{k'_1} \alpha_0 \cdot k'_0$ (which is $(\alpha\,;k')_0$ by definition), and $\pi(e)$ is the path $\alpha_{0*}(F_{v_0}k_0) = (\mathsf{ap}_{k'_1}\alpha_0)^{-1} \cdot F_{v_0}k_0 = (\mathsf{ap}_{k'_1}\alpha_0)^{-1} \cdot (\mathsf{ap}_{k'_1}\alpha_0 \cdot k'_0) = k'_0$.

Proof. (i) Precomposition with p is an equivalence $(V \to V') \simeq (\mathrm{Fib}k \to V')$: the fibres of p are the P_x, which are 2-connected, and V' is a 2-type. (ii) Hence $(a_1, F) \mapsto ((x,q) \mapsto (a_1 x, F_x q))$ is an equivalence

$$\textstyle\sum_{a_1 : V \to V'} \prod_x P_x \to (k'_1(a_1 x) = y'_0) \ \simeq\ (\mathrm{Fib}k \to \mathrm{Fib}k'),$$

by currying, (i), and the distributivity of Π over Σ. (iii) A pointed map is a map with a path $G(w_0) = w'_0$; along (ii) this becomes the path $(a_1 v_0, F_{v_0} k_0) = w'_0$, by definition. (iv) Paths in a Σ-type: for fixed (a_1, F), $(\alpha_0, e) \mapsto (\alpha_0, \pi(e))$ is an equivalence onto $(a_1 v_0, F_{v_0} k_0) = w'_0$, since π composes ap of left multiplication by a fixed path (an equivalence, so its ap is one) with a left and a right multiplication by fixed paths. Reordering the components gives Dec. □

2.2. The key lemma.

For $\beta : \mathrm{Map}_*(Y,Y')$ write $G^\beta_y(q) = \mathsf{ap}_{\beta_1} q \cdot \beta_0 : \beta_1 y = y'_0$ for $q : y = y_0$ (so $(k\,;\beta)_0 = G^\beta_{k_1 v_0}(k_0)$), and $Z^\beta_x = (c_1 x = \beta_1(k_1 x))$. Define

$$\Phi_c : S(c) \to T(c), \qquad \Phi_c(\beta, s) = \big(x\, q \mapsto h s_x \cdot G^\beta_{k_1 x}(q),\ c(s)^{-1}\big),$$

where hs is the homotopy of s and $c(s) : c_0 = h s_{v_0} \cdot (k\,;\beta)_0$ its coherence.

Lemma 6 (key lemma)**.** *For every pointed $c : V \to_* Y'$, Φ_c is an equivalence: pointed null-homotopies of $p\,;c$ are the same as factorisations of c through k.*

For $F_x : P_x \to Q_x^c$ say that F_x is a *translate* of G^β if we are given

$$\mathsf{Tr}_x^\beta(F_x) = \sum_{z : Z_x^\beta} \big((q \mapsto z \cdot G^\beta_{k_1 x} q) = F_x\big).$$

Lemma 7 (paths of pointed maps)**.** *For pointed $c, g : X \to_* Y$, sending a path to its homotopy and coherence is an equivalence $(c = g) \simeq \sum_{h : \prod_x} {}_{c_1 x = g_1 x}(c_0 = h_{x_0} \cdot g_0)$.*

Proof. Fundamental theorem of identity types: the total space $\sum_g \sum_h (c_0 = h_{x_0} \cdot g_0)$ is contractible (first a singleton of c_1 by function extensionality, then a singleton of c_0), and the map agrees with the transport map by path induction. □

Lemma 8 (translates)**.** *For each x, $z \mapsto (q \mapsto z \cdot G^\beta q,\ (z, \mathsf{refl}))$ is an equivalence $Z_x^\beta \simeq \sum_{F_x} \mathsf{Tr}_x^\beta(F_x)$, and each $\mathsf{Tr}_x^\beta(F_x)$ is a proposition.*

Proof. The inverse forgets F_x and the path; one round trip is trivial, the other is path induction. $\mathsf{Tr}_x^\beta(F_x)$ is the fibre over F_x of $\tau : z \mapsto (q \mapsto z \cdot G^\beta q)$, which is the constant map $Z_x^\beta \to (P_x \to Z_x^\beta)$ followed by postcomposition with the fibrewise equivalence $d \mapsto (q \mapsto d\, q \cdot G^\beta q)$ (right multiplication). The constant map is an embedding because P_x is 2-connected and Z_x^β a 3-type (maps from a 2-connected type into the path types, which are 2-types, are constant); so τ is an embedding and its fibres are propositions. □

Lemma 9. *For fixed β, the map $(c = k\,;\beta) \to \sum_{(F,e) : T(c)} \prod_x \mathsf{Tr}_x^\beta(F_x)$, $s \mapsto (\Phi_c(\beta, s),\ x \mapsto (hs_x, \mathsf{refl}))$, is an equivalence.*

Proof. Lemma 7 gives $(c = k\,;\beta) \simeq \sum_{h : \prod_x Z_x^\beta} (c_0 = h_{v_0} \cdot G^\beta_{k_1 v_0} k_0)$. Lemma 8 under $\prod_x$ and distributivity give $\prod_x Z_x^\beta \simeq \sum_F \prod_x \mathsf{Tr}_x^\beta(F_x)$, with $F_{v_0} k_0 = h_{v_0} \cdot G^\beta_{k_1 v_0} k_0$ by definition; the pointing condition is carried along, and inverted. □

The restriction of $(F, e) : T(c)$ to the fibre over the base point, conjugated by c_0, is the pointed map

$$\mathrm{res}(F, e) : \Omega Y \to_* \Omega Y', \qquad q \mapsto c_0^{-1} \cdot F_{v_0}(k_0 \cdot q),$$

pointed by $c_0^{-1} \cdot F_{v_0} k_0 = c_0^{-1} \cdot c_0 = \mathsf{refl}$.

Lemma 10. *For $(F, e) : T(c)$ and $\beta : \mathrm{Map}_*(Y, Y')$, the propositions $\mathsf{Tr}_{v_0}^\beta(F_{v_0})$ and $\mathrm{res}(F, e) = \mathsf{loop}\,\beta$ are logically equivalent.*

Proof. The second is a proposition since $\mathrm{Map}_*(\Omega Y, \Omega Y')$ is a set (ΩY is 2-connected, $\Omega Y'$ a 3-type). We use $\mathsf{loop}\,\beta(q) = \beta_0^{-1} \cdot (\mathsf{ap}_{\beta_1} q \cdot \beta_0)$.

($\Rightarrow$) Given (z, E), $F_{v_0} w = z \cdot (\mathsf{ap}_{\beta_1} w \cdot \beta_0)$ for all w, and with e, $c_0 = z \cdot (\mathsf{ap}_{\beta_1} k_0 \cdot \beta_0)$. Since $\Omega^2 Y'$ is connected, pointed maps $\Omega Y \to_* \Omega Y'$ with equal underlying maps are merely equal; as the goal is a proposition it suffices to check, for all q,

$$(z \cdot (\mathsf{ap}_{\beta_1} k_0 \cdot \beta_0))^{-1} \cdot \big(z \cdot (\mathsf{ap}_{\beta_1}(k_0 \cdot q) \cdot \beta_0)\big) = \beta_0^{-1} \cdot (\mathsf{ap}_{\beta_1} q \cdot \beta_0).$$

This follows from $(z \cdot A)^{-1} \cdot (z \cdot B) = A^{-1} \cdot B$, $\mathsf{ap}_{\beta_1}(k_0 \cdot q) = \mathsf{ap}_{\beta_1} k_0 \cdot \mathsf{ap}_{\beta_1} q$, and $(A \cdot b)^{-1} \cdot ((A \cdot M) \cdot b) = b^{-1} \cdot (M \cdot b)$ (path induction on z and on b).

($\Leftarrow$) Given $r : \mathrm{res}(F, e) = \mathsf{loop}\,\beta$, take $z = F_{v_0} k_0 \cdot (\mathsf{ap}_{\beta_1} k_0 \cdot \beta_0)^{-1}$. For $w = k_0 \cdot q$ we get $F_{v_0} w = c_0 \cdot \mathrm{res}(F, e)(q) = c_0 \cdot \mathsf{loop}\,\beta(q)$, and with e the same identities give $z \cdot (\mathsf{ap}_{\beta_1} w \cdot \beta_0) = F_{v_0} w$. Every w is of this form ($w = k_0 \cdot (k_0^{-1} \cdot w)$); function extensionality gives E. □

Lemma 11. *For every $(F, e) : T(c)$, the type $\sum_\beta \prod_x \mathsf{Tr}_x^\beta(F_x)$ is contractible.*

Proof. For each β, $\prod_x \mathsf{Tr}_x^\beta(F_x)$ and $\mathsf{Tr}_{v_0}^\beta(F_{v_0})$ are logically equivalent propositions: one way by evaluation, the other because a family of propositions over the connected V holding at v_0 holds everywhere. With Lemma 10, the type is equivalent to $\sum_\beta(\mathrm{res}(F,e) = \mathsf{loop}\,\beta)$, the type of deloopings of $\mathrm{res}(F,e)$. It is contractible by Wärn's delooping theorem: since Y is 2-connected and Y' a 4-type, every pointed map $\Omega Y \to_* \Omega Y'$ has a unique delooping. □

Proof of Lemma 6. Φ_c is the composite

$$S(c) \simeq \sum_\beta \sum_{(F,e)} \prod_x \mathsf{Tr}_x^\beta(F_x) \simeq \sum_{(F,e)} \sum_\beta \prod_x \mathsf{Tr}_x^\beta(F_x) \longrightarrow T(c)$$

of Lemma 9 under $\sum_\beta$, a swap, and the first projection, which is an equivalence by Lemma 11. □

Remark 12. A fibrewise map $F_x : P_x \to Q_x^c$ between two "$K(B,3)$-torsors" is a linear part and a translation. The linear part is the delooping β of the restriction over v_0; the translation is the homotopy component, kept as data and fixed at v_0 by the pointing. No point is chosen in any other fibre, no representative of a class is chosen, and lifts are not assumed unique.

2.3. **Coefficients and assembly.**

Lemma 13. *The maps* $a \mapsto K(a,2)$, $\mathrm{Hom}(A,A') \to \mathrm{Map}_*(V,V')$, *and* $b \mapsto K(b,4)$, $\mathrm{Hom}(B,B') \to \mathrm{Map}_*(Y,Y')$, *are equivalences.*

Proof. At level 2 this is the known classification of maps $K(A,2) \to_* K(A',2)$. At level 4, a map G is $K(\mathsf{inv}(G),4)$ by climbing from level 2 with unique deloopings, and $\mathsf{inv}(K(b,4)) = b$ because the looping of $K(b,m+1)$ is the conjugate of $K(b,m)$ by the equivalences $\Omega K(-,m+1) \simeq K(-,m)$, and conjugation is invertible. □

Proof of Theorem 3. ρ is the composite

$$\begin{aligned}\mathsf{Mor}(t,t') &\simeq \textstyle\sum_a S(K(a,2)\,;k') \xrightarrow{\sum_a \Phi} \sum_a T(K(a,2)\,;k') \\ &\simeq \textstyle\sum_\alpha T(\alpha\,;k') \xrightarrow{\mathrm{Dec}} \mathrm{Map}_*(\mathrm{Fib}k, \mathrm{Fib}k'),\end{aligned}$$

using Lemma 13 at the first and third steps and Lemmas 6 and 5. That the composite is the canonical map of Definition 2 holds by definition, including the pointing. □

Remark 14. The canonical map lies over $K(a,2)$, $\rho(a,b,H)\,;p' = p\,;K(a,2)$, and its restriction to the fibre over the base point is $\mathsf{loop}\,K(b,4)$ (Lemma 10). That b is the map induced on π_3 under $\pi_3(\mathrm{Fib}k) \cong B$ is not proved here.

3. The homotopy precategory of triples

Definition 15. For $m = (a,b,H) : t \to t'$ and $m' = (a',b',H') : t' \to t''$ let $m\,;m' = (a\,;a',\ b\,;b',\ H'')$, with H'' the pasting

$$K(a;a',2);k'' = K(a,2);K(a',2);k'' = K(a,2);k';K(b',4) = k;K(b,4);K(b',4) = k;K(b;b',4),$$

built from $K(g;h,n) = K(g,n);K(h,n)$, associativity, H' whiskered by $K(a,2)$, and H whiskered by $K(b',4)$. The identity is $(1_A, 1_B, H_1)$, where H_1 comes from $K(1,n) = \mathrm{id}$ and the unit laws.

The identity $K(1,n) = \mathrm{id}$ ($n = 2,4$) follows from surjectivity of $K(-,n)$: if $K(h,n) = \mathrm{id}$, then $K(1,n) = K(1,n)\,;K(h,n) = K(1\,;h,n) = K(h,n) = \mathrm{id}$.

Proposition 16. $\rho(\mathrm{id}_t)$ *and the identity, and* $\rho(m\,;m')$ *and* $\rho(m)\,;\rho(m')$, *are pointwise equal; hence they are equal as homotopy classes of pointed maps.*

Proof. For a path r of pointed maps let $hp(r)$ be its underlying homotopy. By path induction, hp takes composites to composites and inverses to inverses, $hp(\mathsf{ap}_{u;-}s)_x = hp(s)_{u_1 x}$, $hp(\mathsf{ap}_{-;v}s)_x = \mathsf{ap}_{v_1}(hp(s)_x)$, the associativity path has trivial homotopy, and so has a path changing only the pointing. Then: (1) the canonical map of a square transported along paths $u_2 = u$, $v_2 = v$ is pointwise the canonical map of the square (path induction on the two paths); (2) for a pasted square, the homotopy is $hs'_{u_1 x} \cdot \mathsf{ap}_{v'_1}(hs_x)$, and the second components agree by ap of composites and reassociation; (3) the canonical map of the identity square is pointwise the identity. Pointwise equality gives mere equality of pointed maps, because the loop space of the target is connected. □

No coherent pointed path is constructed; the homotopy precategory only needs classes.

Theorem 17. *Let* $\hom(t,t') = \|\mathsf{Mor}(t,t')\|_0$*, with the composition and identities induced by the definition above. This is a precategory, and the functor which is* realise *on objects and the class of* ρ *on hom-sets, into the homotopy precategory of pointed simply connected* 3*-types, is an* isomorphism of precategories*: an equivalence of object types and a bijection on hom-sets, preserving identities and composition.*

Proof. The class of ρ is a bijection by Theorem 3, and realise is an equivalence of types. By Proposition 16 it preserves composition and identities, and the category laws in hom follow from those of pointed maps up to homotopy through the bijections. □

Remark 18. Following Ahrens, Kapulkin and Shulman (*Univalent categories and the Rezk completion*, Definition 6.9), a precategory may have an arbitrary type of objects, here 2-types, while its hom-types are sets. Neither precategory is univalent, and no such claim is made. By their Lemma 6.16, under univalence an isomorphism of precategories yields an equality of precategories.

4. The components of the compatible homotopies

The sphere arguments compute $\hom(t_0,t')$ directly from $\mathsf{Mor}(t_0,t')$. The coefficient b is fixed by cohomological information about $K(\mathbb{Z},2)$, and the components of the type of homotopies by (H3). The latter concerns loops at an arbitrary point of $\mathrm{Map}_*(K(\mathbb{Z},2),K(B',4))$, which we move to the base point.

Lemma 19 (translation)**.** *Let* X*,* Z *be pointed,* $M = \mathrm{Map}_*(X,\Omega Z)$ *and* $G : M$*. Pointwise right multiplication by* G^{-1} *is an equivalence* $M \to M$ *sending* G *to a map merely equal to the constant map, when* $\Omega^2 Z$ *is connected.*

Lemma 20 (components)**.** *Let* $Y \simeq_* \Omega Z$ *with* $\Omega^2 Z$ *connected, and suppose every pointed map* $X \to_* \Omega^2 Z$ *is merely constant. Then any two paths* $f = g$ *in* $\mathrm{Map}_*(X,Y)$ *are merely equal.*

Proof. Transport along $Y \simeq_* \Omega Z$ and translate by g^{-1} (Lemma 19); this reduces to two paths $r, r' : u = v$ with v merely, hence (the goal being a proposition) actually, the constant map. Then $r^{-1} \cdot r'$ is a loop at the constant map, that is, a pointed map $X \to_* \Omega^2 Z$, merely constant by hypothesis. □

For $X = K(\mathbb{Z},2)$ and $Y = K(B',4) \simeq_* \Omega K(B',5)$, the hypothesis is (H3): $[K(\mathbb{Z},2),K(B',3)]_*$ is a point, since $\Omega^2 K(B',5) \simeq_* K(B',3)$.

5. $\pi_3(S^2)$ through triple morphisms

Let $\kappa : K(\mathbb{Z},2) \to_* K(\mathbb{Z},4)$ be the cup square, $t_0 = (\mathbb{Z},\mathbb{Z},\kappa)$ and $P = \mathrm{Fib}\kappa$. We use:

- (H4): $b \mapsto [\kappa\,;K(b,4)]$ is a bijection $\mathrm{Hom}(\mathbb{Z},B') \to [K(\mathbb{Z},2),K(B',4)]_*$;

- (H3): $[K(\mathbb{Z},2), K(B',3)]_*$ is a point.

Proposition 21. *For every triple $t' = (A', B', k')$, the map $\theta : \hom(t_0, t') \to \mathrm{Hom}(\mathbb{Z}, A')$, $[a, b, H] \mapsto a$, is a bijection.*

Proof. Surjective: for a, (H4) gives b with $[\kappa\,;K(b,4)] = [K(a,2)\,;k']$, hence merely a homotopy H. *Injective:* if two morphisms lie over a, their coefficients b, b' have the same image under (H4), so $b = b'$, and their homotopies are merely equal by Lemma 20. □

Proposition 22. *Let $s \in [S^2, P]_*$ be the class with $s\,;[p] = [\sigma]$, where $\sigma : S^2 \to_* K(\mathbb{Z},2)$ is the generator. For every triple t', $s^* : [P, \mathrm{Fib}k']_* \to [S^2, \mathrm{Fib}k']_*$ is a bijection.*

Proof. The composite

$$\hom(t_0, t') \xrightarrow{[\rho]} [P, \mathrm{Fib}k']_* \xrightarrow{s^*} [S^2, \mathrm{Fib}k']_* \xrightarrow{p'_*} [S^2, K(A',2)]_*$$

sends $[m]$ to $[\sigma\,;K(a,2)]$, because $\rho(m)$ lies over $K(a,2)$. The map $a \mapsto [\sigma\,;K(a,2)]$ is a bijection $\mathrm{Hom}(\mathbb{Z}, A') \to [S^2, K(A',2)]_*$: maps $S^2 \to K(A',2)$ correspond to maps $K(\mathbb{Z},2) \to K(A',2)$ through σ, and these to homomorphisms. So the composite is a bijection. The first map is a bijection by Theorem 3, and p'_* by lifting from the source S^2 (maps from S^2 into $\mathrm{Fib}k'$ correspond to maps into $K(A',2)$). Hence s^* is a bijection. □

Corollary 23. *$\pi_3(S^2) \cong \mathbb{Z}$, from the universal property of S^2.*

Proof. By the classification of objects, every pointed simply connected 3-type is $\mathrm{Fib}k'$ for some triple, so s^* is a bijection for all of them. Then P is the 3-truncation of S^2 (a two-map Yoneda argument), and $\pi_3(S^2) \cong \pi_3(P) \cong \mathbb{Z}$. □

6. $\pi_4(S^3)$ through triple morphisms and the looping embedding

Looping is an embedding of pointed 2-connected 4-types into pointed simply connected 3-types; its image is the *deloopable* triples. On maps, looping is a bijection $[X, Y]_* \simeq [\Omega X, \Omega Y]_*$ for X 2-connected and Y a 4-type. Let $\bar{k} = \kappa\,;K(\rho,4) : K(\mathbb{Z},2) \to_* K(\mathbb{Z}/2,4)$ be the cup square reduced mod 2, $\bar{t} = (\mathbb{Z}, \mathbb{Z}/2, \bar{k})$, $R = \mathrm{Fib}\bar{k}$, and BR its delooping. The universal property to establish is statement M: for a marked class $s \in [S^3, BR]_*$, the map $s^* : [BR, Y]_* \to [S^3, Y]_*$ is a bijection for every pointed 2-connected 4-type Y. By the bijection on maps, M at Y is equivalent to bijectivity of $r^* : [R, \Omega Y]_* \to [\Omega S^3, \Omega Y]_*$ for the looped class r.

Proposition 24. *For every deloopable triple $t' = (A', B', k')$, the map $\theta : \hom(\bar{t}, t') \to \mathrm{Hom}(\mathbb{Z}, A')$ is a bijection.*

Proof. Surjective: by (H4), $[K(a,2);k'] = [\kappa;K(h,4)]$ for some h. Since t' is deloopable, $2h(1) = 0$ (Statement T, the restricted 2-torsion lemma). So $h = \rho\,;g$, and $[K(a,2)\,;k'] = [\bar{k}\,;K(g,4)]$. *Injective:* $g \mapsto [\bar{k}\,;K(g,4)]$ is injective, which fixes b; the homotopies are merely equal by Lemma 20 and (H3). □

Proposition 25. *Statement M holds for every pointed 2-connected 4-type Y.*

Proof. Let t' be the triple of ΩY; it is deloopable.

Lifting from ΩS^3. A map $F : \Omega S^3 \to_* K(A',2)$ is, up to homotopy, $r\,;\bar{p}\,;K(a,2)$ for some a. This uses that r^* is bijective into $K(A',2)$ (M at $K(A',3)$, looped), that $\bar{p}^*$ is bijective on maps into 2-types, and Lemma 13. By Proposition 24 there is merely a morphism (a, b, H), so $F\,;k'$ is merely $r\,;\bar{p}\,;\bar{k}\,;K(b,4)$, which is merely constant. Together with uniqueness of lifts from ΩS^3, maps $\Omega S^3 \to \mathrm{Fib}k'$ correspond to maps $\Omega S^3 \to K(A',2)$.

The square. The composite

$$\hom(\bar{t}, t') \to [R, \mathrm{Fib}k']_* \xrightarrow{r^*} [\Omega S^3, \mathrm{Fib}k']_* \to [\Omega S^3, K(A',2)]_*$$

sends $[m]$ to $r;[\bar{p};K(a,2)]$, and $a \mapsto r;[\bar{p};K(a,2)]$ is a bijection. So r^* is bijective at $\mathrm{Fib}k' \simeq \Omega Y$, and M holds at Y. □

Corollary 26. *$\pi_4(S^3) \cong \mathbb{Z}/2$, from the universal property of S^3.*

Proof. Apply statement M at $Y = BR$ and at $Y = \|S^3\|_4$; the two-map Yoneda argument gives $\pi_4(S^3) \cong \pi_4(BR) \cong \pi_3(R) \cong \mathbb{Z}/2$. □

7. What is used

The sphere arguments rest on Theorem 3 and the computation of hom-sets from Mor. They reuse the following:

- the cup square and its normalisation;
- (H3) and (H4) for $K(\mathbb{Z},2)$;
- the restricted 2-torsion lemma;
- Wärn's delooping theorem and the looping embedding;
- the elementary computations of maps out of S^2, S^3 and ΩS^3, together with the marked classes.

They do not use any earlier, specialised mapping theorem for the targets $\mathrm{Fib}k$: this was confirmed formally by deleting those results from a copy of the development, which still checks.